Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

<think>

the de Broglie wavelength of a particle is given by the formula Î» =h /p , where Î» is the wavelength , h is the Planck's constant , and p is the momentum of the particle .

the momentum of a particle is given by the formula p = âĪļ (2 *m * E), where m is the mass of the particle and E is the kinetic energy .

we are given that the kinetic energy is 1 eV , which is equal to 1 . 6 ÃĹ 10 ^- 19 J .

the mass of an electron is approximately 9 . 11 ÃĹ 10 ^- 31 kg .

first , we convert the kinetic energy from eV to Joules : 1 eV = 1 . 6 ÃĹ 10 ^- 19 J .

the momentum is given by p = âĪļ (2 * 9 . 11 ÃĹ 10 ^- 31 kg * 1 . 6 ÃĹ 10 ^- 19 J).

then , we can calculate the wavelength by dividing the Planck's constant (6 . 626 ÃĹ 10 ^- 34 m ^ 2 kg / s) by the momentum .

let's perform the calculation and convert to cm .

after the calculation , we obtain a value of 1 . 227 ÃĹ 10 ^- 7 cm .

</think><answer> G </answer>